

Deep Drift v1.4: A smooth start, a living pause, a new dive every time

Report for Hasib, 25 September 2026. What is on the website now, what changed in this version, and what to do next.

In one sentence: Five things Hasib noticed while testing v1.3 are fixed: the blink and lag at the start, jerky keyboard turning, the same animals every dive, a pause that froze the whole sea, and school fish that looked too slow.

1. What changed in v1.4

1. The camera blinks and lags at the beginning.

What we found: Measured on this laptop's own graphics (Intel UHD) at its screen size (1536x864 at 125%). The reef, which is the start of the dive, took 50 to 70 ms per picture (about 15 to 20 pictures a second). After a few seconds the automatic sharpness control lowered the resolution; that change cleared the picture for one frame (a blink) and caused a 170 to 200 ms hitch. A second blink came from Begin itself: the picture cut to black and faded back in.

What we did: Before Begin, the page now draws the busy reef a few times and picks the sharpness that gives about 30 ms per picture, so nothing has to change during the dive. Sharpness changes only after about a second of slow pictures, and never between drawing and showing a picture. Edge smoothing is off on scaled screens (it cost about a fifth of each picture there). The rock grain skips detail you cannot see at a distance; coral shapes share their corners so the graphics card does less work. The loading screen no longer freezes for long. Begin now fades the intro card away over the live picture.

Result: On this laptop: the first 8 to 12 s of the dive run at a steady 60 pictures a second with no hitch over 33 ms (v1.3: 200 ms), no resolution change during the dive, and the longest freeze while loading is 0.37 s (v1.3: 1.85 s). The price: on this laptop the picture is drawn at about 0.86 to 1.0 of the screen's sharpness, so it is a little softer.

2. Keyboard rotations are not smooth.

What we found: Each arrow key press jumped the view by 7 degrees, and holding the key repeated those jumps in steps after a half-second delay.

What we did: Holding an arrow key now turns the head smoothly: it speeds up to a steady turn (about 75 degrees a second) and eases to a stop when let go, like turning your head.

Result: Measured: the biggest change from one picture to the next is 0.14 degrees, and the turn stops smoothly.

3. Animals and fish play the same all the time; it could be randomised so it does not become boring.

What we found: All the scripted animals (the turtle, the manta, the sharks, the whales, the squid, the anglerfish and more) were placed by hand at fixed times and places.

What we did: Every dive, and every restart, rolls a new version: each scripted animal comes a few seconds earlier or later, nearer or farther, from either side, a little faster or slower, and about one dive in seven it is not there. Circling sharks, grenadiers and schools start anywhere on their circle and may circle the other way; jellyfish drift from new spots; the dolphins still travel as one pod. Every version is checked so no animal swims through rock; if no version is clear, that animal is simply not in that dive.

Result: 73 of 74 scripted animals change between dives; 1 to 7 are absent in a given dive; 0 paths through rock within 50 m of the dive path. The check also found three older mistakes and fixed them: the three reef sharks circled through the reef wall, and the manta met you inside the wall (it now glides over you).

4. The pause button freezes everything; the diver could be frozen while other movement goes on.

What we found: Everything in the sea moved by the dive's clock, so pausing the dive stopped the whole world.

What we did: Pause now stops only the diver: you hover in place. The sea has its own clock: schools, circling sharks and jellyfish keep swimming, animals passing by swim on (when you continue they have moved on, as in real life), and new visitors still come past. Your breathing, bubbles and gentle sway go on.

Result: Measured over 3 s of pause: the diver moved 0.08 m (the breathing bob), the dive time did not move, and all 8 nearby schools and all 12 nearby animals kept moving.

5. Schools of fish do not swim that slowly in real life; check this and give a real-life feeling.

What we found: Checked against research: fish cruise at about 1 to 3 body lengths a second, and each tail beat moves a fish about 0.7 of its body length (Bainbridge 1958). Our 18 cm reef fish already moved 1.1 to 1.4 m/s but beat their tails only about twice a second, so they looked like they were gliding in slow motion. Each school also circled as one rigid block at a constant speed.

What we did: Small schooling fish now beat their tails about 8 times a second (lanternfish and hatchetfish about 5, the barramundi about 2, big animals unchanged: a whale shark beats once every few seconds). Schools surge: their speed rises and falls by about 40%. Each fish weaves a little on its own and faces the way it really moves.

Result: Measured on a reef school over 40 s: its speed ranged from about half to about twice its average. Source: R. Bainbridge (1958), The speed of swimming of fish as related to size and to the frequency and amplitude of the tail beat, *Journal of Experimental Biology* 35: 109-133, <https://journals.biologists.com/jeb/article/35/1/109/13233>.

2. Tests

Opening, on this laptop's graphics (tests/startup.py --gpu)	Begin ready in 3.4 s; first 8 s of the dive at 60 pictures a second, no hitch over 17 ms; 0 shaders built late
Solid world and no pops (tests/solid.py)	Whole dive plus reef and wreck runs: 0 times inside something solid, 0 animal or fish overlaps, 0 pops, 0 jumps; whale shark pushes the diver, a jellyfish is nudged away
Pause, variety, keys, schools (tests/life.py, new)	All pass (numbers above)
Swimming (tests/glide.py)	Steers, never rises, sinks slower looking up and faster looking down, roaming limit works
Menus on phone and desktop (tests/ui.py), live play (tests/live.py)	Pass, no page errors

3. Pictures

Start of the dive on Hasib's laptop (Intel UHD graphics, 1536x864 at 125%): v1.3 (grey) and v1.4 (blue)

Longest hitch in the first 10 to 12 s of the dive (shorter is better)



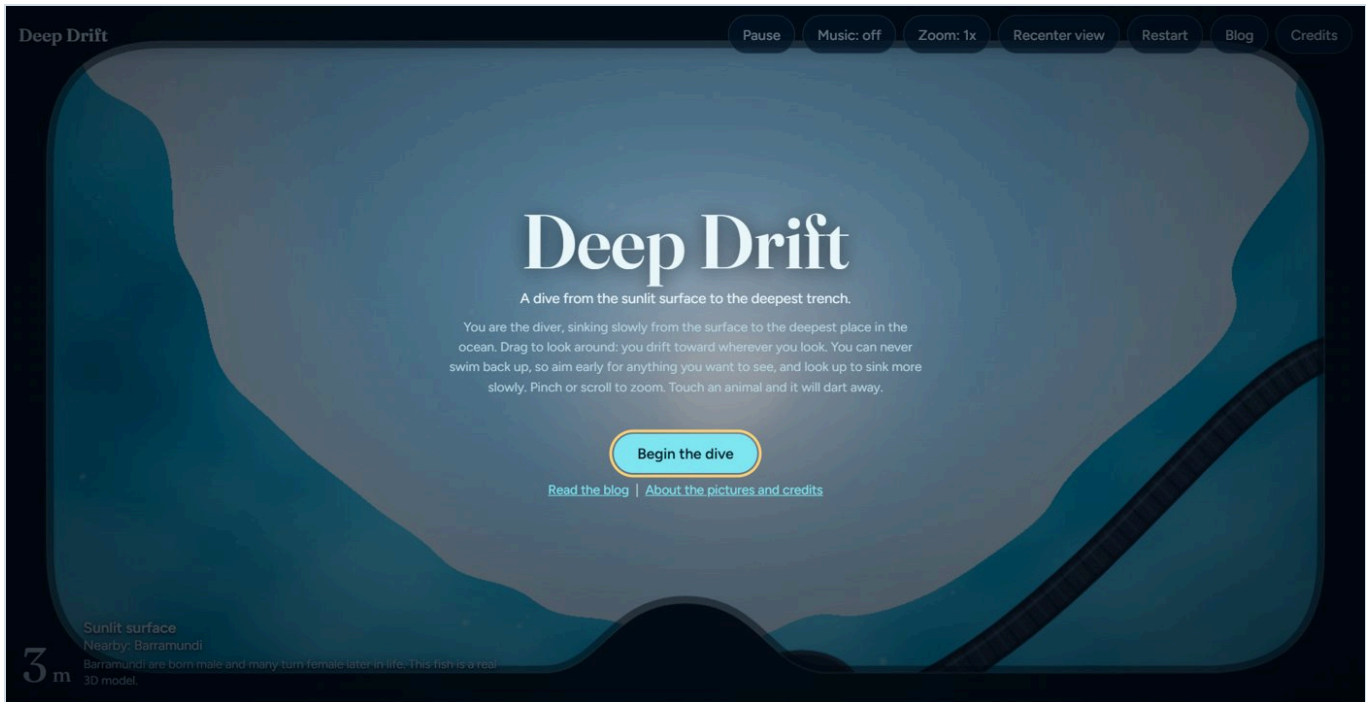
Time for one picture of the busy reef (full sharpness 1.25) (shorter is better)



Loading screen: longest freeze (shorter is better)



The start of the dive on Hasib's laptop, v1.3 and v1.4.



The start page (the loading message counts up; Begin fades this card away).



The dive on Hasib's laptop at its screen size: reef at 6, 15 and 34 m, the drop-off at 155 m, the vents at 1,575 m, the abyssal plain at 5,435 m.

4. What is on the website now

Deep Drift is a first-person dive in the browser, from the sunlit surface to the floor of the Challenger Deep (10,935 m), as one continuous descent of about 7 minutes. You are the diver: you drift toward wherever you look and you are always sinking. The places are real kinds of places moved closer together, and the descent is sped up; the page says so.

The dive, top to bottom

Surface and coral reef	0 to 60 m	A boat above you, dolphins passing at the start, a reef shelf and wall covered in corals, sponges, sea fans and anemones. Barramundi schools (the one real 3D model), reef fish schools, blacktip reef sharks, a green sea turtle, a giant manta ray, moon jellies.
Reef drop-off and deep reef	60 to 200 m	The wall continues into the blue: big sea fans and black corals. A whale shark, a great white shark and a humpback whale pass by; a school of silver jacks swirls beside you.
Twilight zone	200 to 1,000 m	Light fades. Jellyfish, hatchetfish, lanternfish, a sperm whale hunting, a siphonophore (a glowing chain of animals), a vampire squid.
Midnight zone	1,000 to 1,600 m	Only your torch and lamp. A giant squid, an anglerfish with its glowing lure, a gulper eel with a glowing tail tip, glowing jellyfish.
Hydrothermal vents	about 1,600 m	Smoking chimneys, giant tube worms, vent shrimp swarms, droplets of liquid CO ₂ .
Volcano slope	1,600 to 4,500 m	Bamboo corals, glass sponges, sea lilies, brittle stars.
Abyssal plain	about 5,000 m	An imagined shipwreck with rusticles, a whale skeleton, manganese nodules, sea pigs, xenophyophores, dumbo octopuses, grenadiers and tripod fish.
Trench and Challenger Deep	6,000 to 10,935 m	The trench wall, Mariana snailfish (only between about 6,500 and 8,300 m, as in reality), sea cucumbers, amphipods, and the deepest floor.

How you control it

Look and swim	Drag with the mouse or a finger. You slowly swim toward where you look. Looking up slows your sinking, looking down speeds it up; you can never swim back up.
Arrow keys	Turn your head smoothly (it speeds up while held and eases to a stop).
Zoom	Pinch, or scroll the mouse wheel, or the Zoom button.
Pause	Button or Space. You hover in place while the sea keeps moving around you. Play to swim on.
Recenter view	Button or double-click: look along the dive again.
Restart	Start the dive again. Each dive is a little different.
Music	A live soundtrack made in the browser.

Phones	The buttons sit in a Menu. Motion look turns the view with the phone.
Blog and Credits	Short posts in our own words with source links; credits say what is real and what is built by code.

How it feels

- Nothing passes through you: corals, rocks, sponges, tube worms, the wreck and the bones are solid, and you slide around them.
- Animals keep a comfortable distance and swim aside; big ones (whales, the whale shark, the manta) gently push you aside.
- School fish part around you and beat their tails as fast as real fish of their size.
- Nothing pops in or out: visitors arrive from out of sight and leave out of sight; far things fade into the water.
- Each dive is a little different: animals come a bit earlier or later, on either side, nearer or farther, and some are not there every time.
- Animals only live where they really live (for example no fish below about 8,300 m).

What is real

Real: the barramundi (a CC0 3D model by Microsoft from the Khronos glTF samples) and the fonts. Everything else (all other animals, corals, rocks, the wreck, the water, the light) is built by code, so it is stylised, not photographic.

Running it

On your computer: in the project folder run `node tools/serve.mjs` (or `npm start`) and open `http://localhost:8080`. Nothing is loaded from other websites.

5. Known limits (honest list)

- On this laptop the reef is drawn a little softer (about 0.86 to 1.0 of full sharpness) to stay smooth. The reef (corals and rock shading) is still the heaviest part.
- Animals are kept off rock and the floor but not off corals, and beyond 50 m from the dive path a straight-swimming animal can still meet rock (hidden by the water and by the rock itself).
- The solid shapes are simple, so you stop a little before thin, airy corals such as fans.
- Deep scenes are lit only by your torch and lamp, so far rock is dark; animals right in front of the torch look paler than they should.
- Nothing has been tried on a real phone yet.

6. What to do next

When	What	Details
v1.5	Animal behaviour	Curious and startled school fish, predators chasing schools, hidden animals (day octopus with ink, flounder, scorpionfish, garden eels), deep-sea light displays, feeding at the vents and the whale skeleton, animals steering around corals.
any time	Phone check	Try the dive on a real phone (speed, motion look, the glide). Needs your phone.

When	What	Details
any time	Lighter reef	Draw far corals with simpler shapes so laptops like this one can use full sharpness.
after testing	Publish	First public version on GitHub Pages (free). Needs a GitHub account.
v1.6+	Real 3D models	In small batches from Sketchfab, each with your OK on its licence and a credit on the page.